Thermodynamics Clements Invitational: Key

Clements High School

12/16/17

1 Thermodynamics

1.1. C

1.2. C

1.3. D

1.4. B

1.5. B

1.6. C

1.7. C

1.8. A

1.9. B

1.10. B

1.11. D

1.12. A

1.13. D

1.14. C

1.15. -40

1.16. -17.78 (accept -17 to -18)

1.17. 4.44 (accept 4 to 5)

1.18. 0

1.19. 100

1.20. -3.9 (accept -3 to -4 if work is correct)

1.21. E

1.22. C

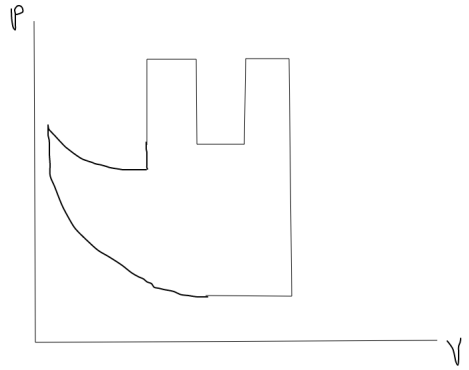
1.23. A

1.24. D

1.25. B

2 A really cool engine!!

(5 points) Relative location of points should be accurate, distances need not be correct.



GRADE WITH LENIENCY-± 5 units for each answer is acceptable in first table and ± 1% of answer is acceptable in second table (except for exact 0 answers)

In the following table, let 1 denote the initial state of the system, and 10 denote the last state of the system before it returns to initial conditions. Fill in all missing variables. 1 point per blank

Variables	1	2	3	4	5	6	7	8	9	10
P(atm)	100.00	100.00	70	70	100.00	100.00	10	10	75.00	45
V(L)	.500	1.00	1.00	1.500	1.500	2.00	2.00	1.00	.3	.500
T(K)	609.3	1218.7	853.06	1279.59	1827.99	2437.3	243.73	121.87	272.827	272.827

Calculate the amount of work done by the system in each process, as well as the amount of heat added or released in each process. 1 point per blank

Step	Work(J)	Heat(J)
1	5066	12622
2	0	-4557
3	3546	8865
4	0	6880
5	5066	12699
6	0	-27356
7	-1013	-2527
8	-1882	0
9	1165	1165
10	0	4237

Efficiency $\eta = 25\%$